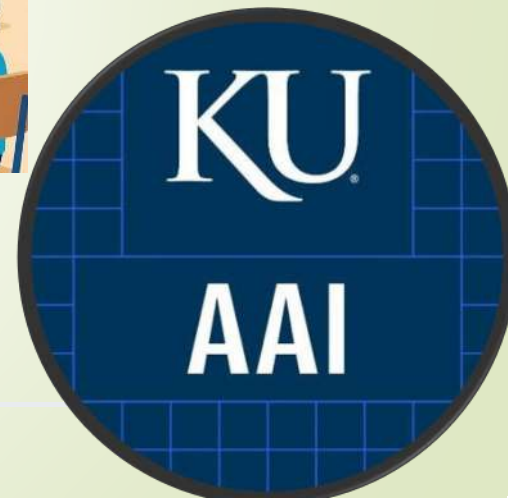


Safe and Responsible AI in K -12 Education

A Safety Protocol and Procedure Guide for Educators



Presentation By:
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MEET YOUR PRESENTER



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Chief AI Consultant · Center for Reimagining Education

Adjunct Professor & Instructional Designer · Johns Hopkins University

25+

Years
Teaching

165+

International
Presentations

\$15M+

Funded
Research

+20

Peer-Reviewed
Publications

2

Books
(2025–2026)

*Host & Co-Creator · AI Advocates Podcast
(listeners in every country · 29 episodes)*



Today's Agenda | 9:30 - 11:00 AM

9:30AM
15 min

Welcome, Framing & Why AI Now

What AI is (and isn't) and the workforce shift driving urgency in K-12.

9:45AM
10 min

The Real Problem in the Classroom

Educator burnout, personalization gap, and AI.

10:00AM
30 min

Safety Protocols & Procedures

Acceptable use policies, vetting, citation, network security, & incident response.

10:30AM
20 min

Build Your Plan (Hands-On)

AI literacy, PD, parent collaboration, then draft your building's immediate action steps.

10:50AM
10 min

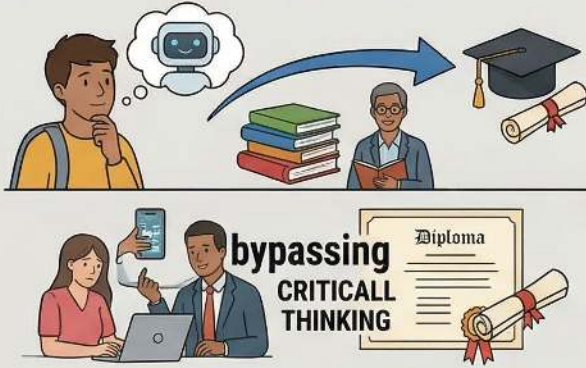
Q&A, Resources & Next Steps

Open questions, national frameworks, guidelines, and where to continue learning.

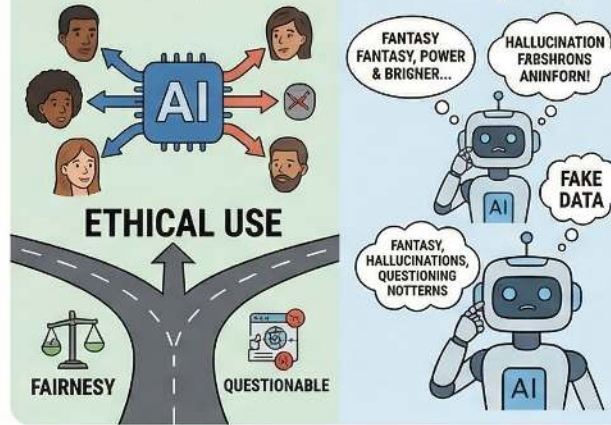
Why AI?

I Choose to Know Not to Fear

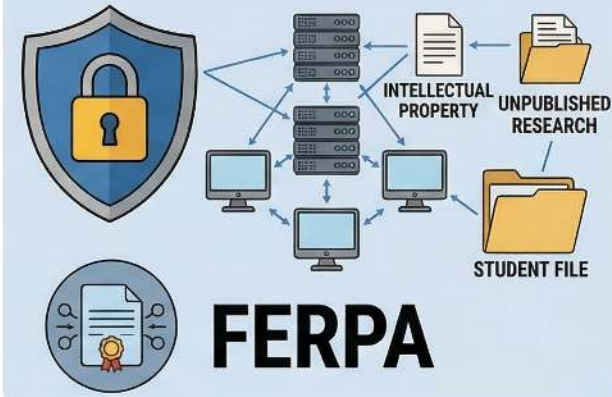
1. Academic Integrity & "The Shortcut"



2. Bias, Ethics, and Equity



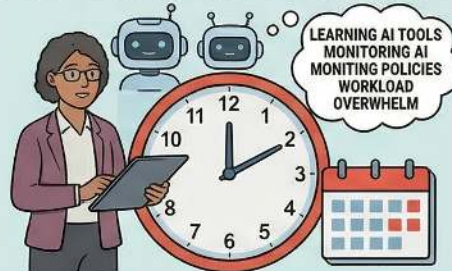
3. Data Privacy and Security



4. Role Displacement & Devaluation



5. The "Time Tax" of Learning



6. Erosion of Human Connection



HUMANITY SOLVES PROBLEMS BY INVENTING NEW TOOLS



HUNTER-GATHERER (50,000-10,000 BCE)

Survival,
fire control.

Est. Tools:
~50-200

Key Impacts:
Humanity:
Oral traditions,
basic shelter.



AGRICULTURAL (10,000 BCE-1750 AD)

Farming,
settlement.

Est. Tools:
~3,000-10,000

Key Impacts:
Humanity:
Cities, writing,
specialization.



INDUSTRIAL (1750-1950)

Electrical energy,
fossil fuels,
mechanization.

Est. Tools:
~500,000-1,000,000

Key Impacts:
Humanity:
Cities, writing,
specialization.



Est. Tools:
~500,000-1,000,000

Key Impacts:
Mass production,
global transport,
urbanization.

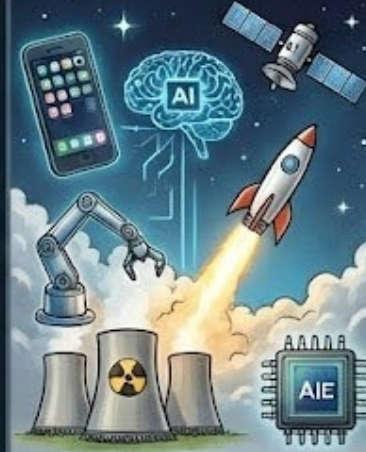


INFORMATION (1950-2000)

Digitization,
computing, global
communication.

Est. Tools:
~1,000,000,000+

Key Impacts:
Humanity:
Internet, software,
data processing.



AI AGE (2000-Future)

Artificial intelligence,
nuclear energy,
space technology.

Est. Tools:
-

Key Impacts:
Humanity:
interplanetary
explored.



Est. Tools:
Unbounded/Generative

Key Impacts:
Automation,
interplanetary exploration,
advanced energy.



IT'S NOT THE TOOLS. IT'S HOW WE USE THEM.

THE PATH TO TRUE AI MASTERY.

SOCRATES:
WRITING WEAKENS
THE MIND.

PLATO:
WRITING IS A TOOL,
BUT DOES NOT HOLD
THE TRUTH.

ARISTOTLE:
WRITING EXPANDS
KNOWLEDGE AND
HUMAN POTENTIAL.



ATROPHIES
MEMORY



CREATES
DEFENSELESS
ARGUMENTS



KILLS THE ACTIVE
DIALOGUE
REQUIRED TO
FIND TRUTH



USE WITH
DISCERNMENT



AMPLIFY
WISDOM

Daisy: "The chimney is not drawing properly,
the oven is not hot enough."

Mrs. Patmore: "It's a poor workman
who blames his tools."

ADAPT
LEARN
COLLABORATE

ELEVATE

HUMAN WISDOM + AI POWER

CANDIDATES EXPECTED TO:

- ✓ Use AI to analyze information
- ✓ Evaluate AI outputs
- ✓ Explain how AI improves workflow quality and speed



WHAT EMPLOYERS ARE ACTUALLY SIGNALING



AI is embedded into the work,
not just the job description



AI competence is often
implicit, not explicitly listed



AI tools are being used in
interviews and evaluated
during hiring tasks



USE AI WISELY, CREATE REAL IMPACT.



Better Decisions



Faster Innovation



Stronger Results

AI Definition & Types

“Intelligence: ability to accomplish complex goals,”

Max Tegmark, Life 3.0: Being Human in the Age of Artificial Intelligence



Devices

Machines

Vehicles

Weapons

Robots

BOTS

Platforms

Agents

Level 1

Level 5

Level 3

The levels of AI progress according to OpenAI:



Level 1: Chatbots, 2.4 Billion Users in 2026



Level 2: Reasoners, human-level problem solving



Level 3: Agents, ~300M Users



Level 4: Innovators, AI that can aid in invention (AlphaFold AlphaProof, cure diseases Isomorphic Drug Design Engine)



Level 5: Organizations, AI that can do the work of an organization



ROI FOCUS

I use AI to accelerate learning loops, allowing me to master new protocols in half the time.



SAFETY FOCUS

I understand the risks of "AI Slop." I use critical thinking to verify every output against company policy.



EFFICIENCY

I use it to automate repetitive structuring so I can focus on high-level problem solving.

An entry level employee knows information but needs help to solve a problem. Level 1:

A good employee can solve a problem without the help of supervision. Level 2:

A great employee solves a problem before others are aware that the problem is even there. Level 3:

Critical Thinking Literacy



Information & Media



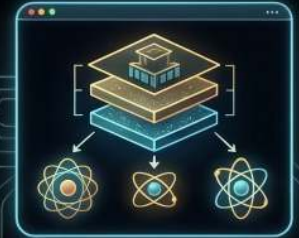
Conceptual
Adaptation



Accessible
Communication



Data & Statistics



First-Principles
Thinking



Meta-Cognition



Logical Argumentation



Systems Thinking



Epistemic Humility

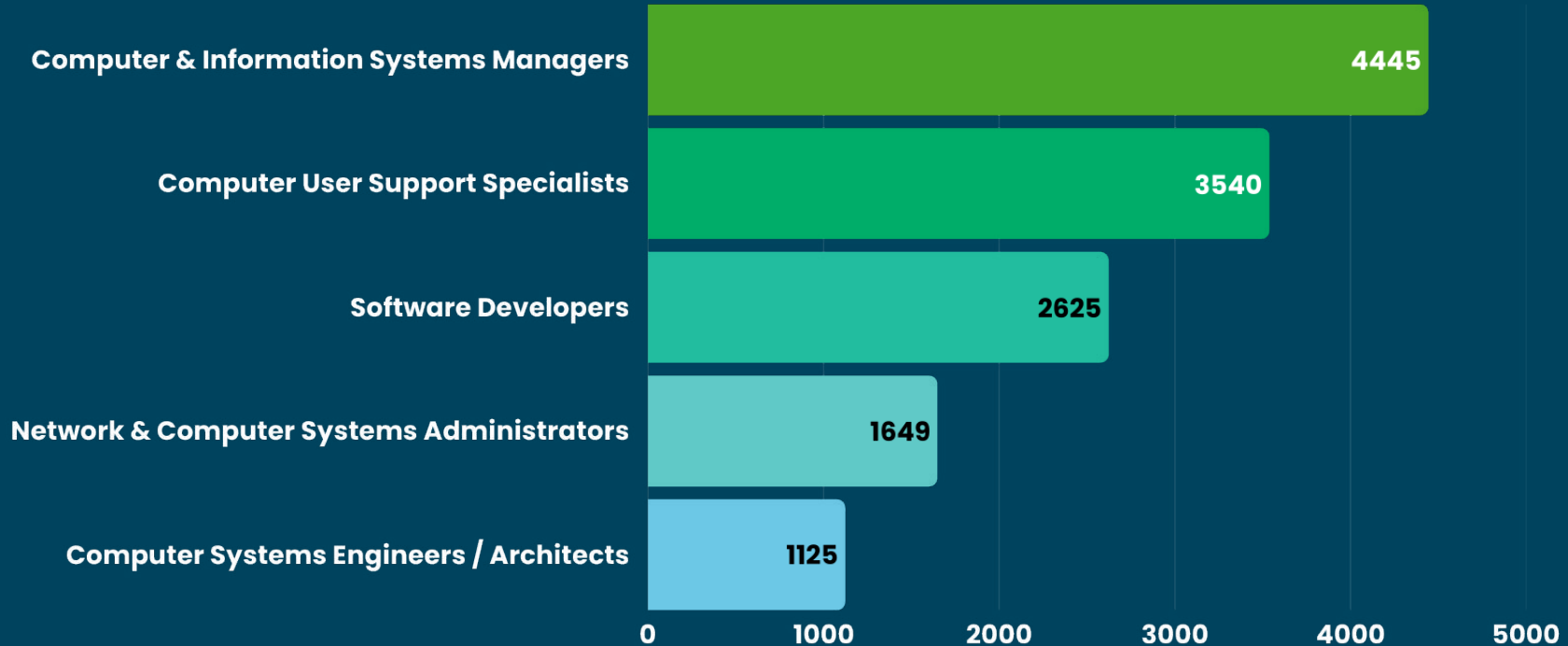


Continuous Learning

The AI Loop: Continuous Adaption



HIGHEST Tech DEMAND JOBS IN 2025



9:45AM But We Can't Forget

The Educator **Burnout Crisis**

Teachers today face unprecedented administrative burdens.

- Endless lesson planning.
- Mountains of grading.
- Data tracking and reporting.

These tasks consume hours of personal time, leaving less energy for what matters most: direct student interaction and meaningful relationship building.



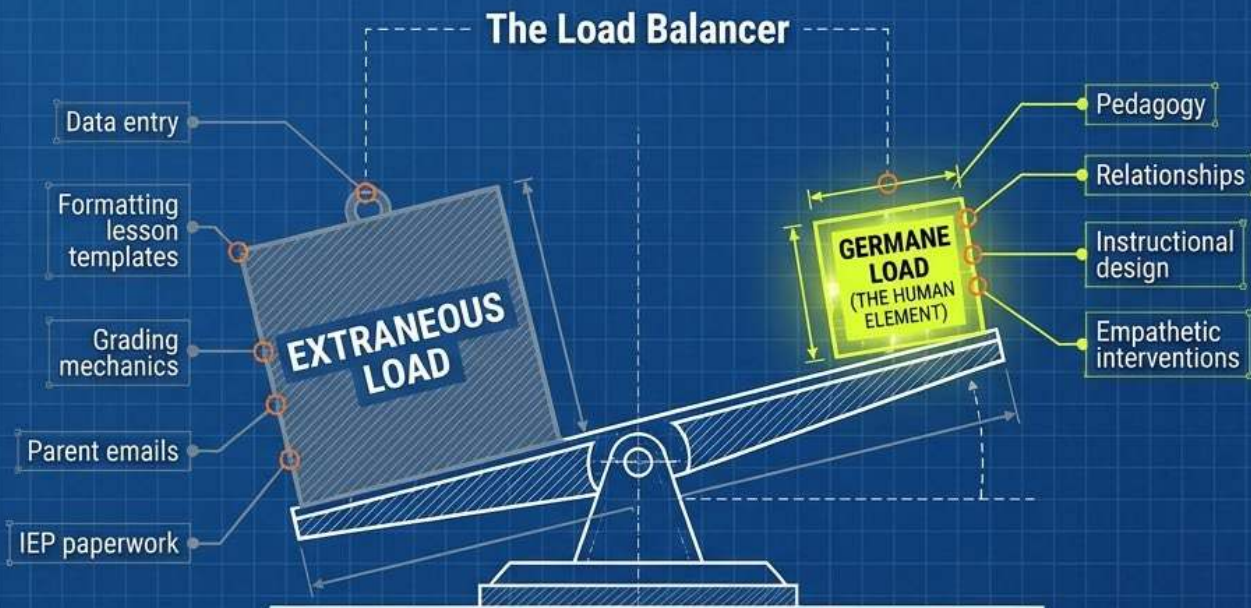
The Real Problem in the Classroom

**50.3
HOURS**

Average teacher workweek

43%

Time spent actively teaching
(DfE Workload Survey, 2024)



You are **exhausted** because you are doing the **heavy lifting of logistics**, not just the heavy lifting of teaching.

Personalized Learning for Every Student — Without Burnout



ELL

AI translates materials,
creates bilingual scaffolds &
vocabulary support



IEP / 504

Differentiates reading levels,
formats text for accessibility,
modifies assessments



Gifted

Generates extension
projects, higher-order
questions & personal paths



At-Risk or Struggling

Creates graphic organizers,
audio versions & simplified
summaries instantly



Absent

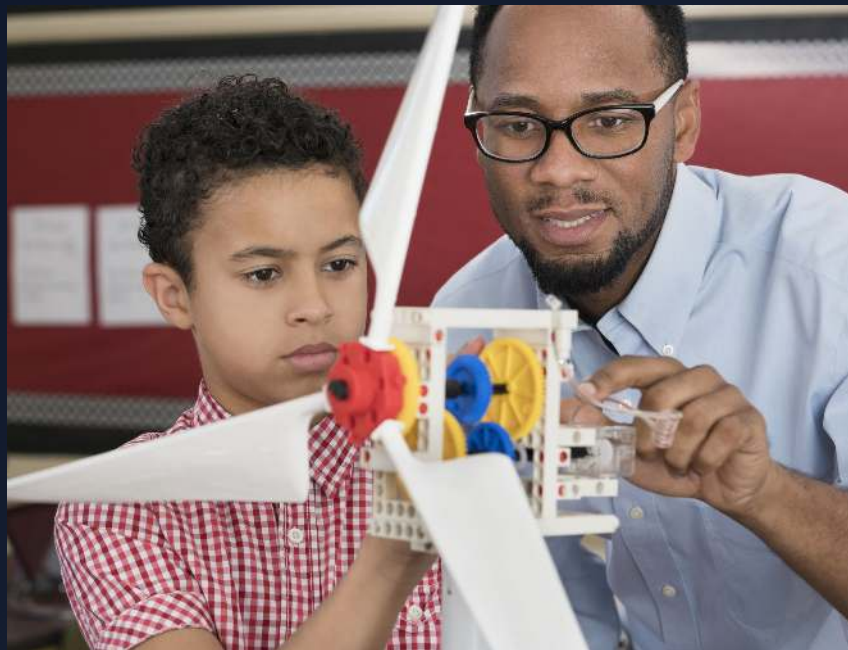
Produces catch-up packets,
summarize missed
instruction & check-ins



Every Student

Formative feedback loops,
personalized practice sets &
progress tracking

Reaching **Every** Student

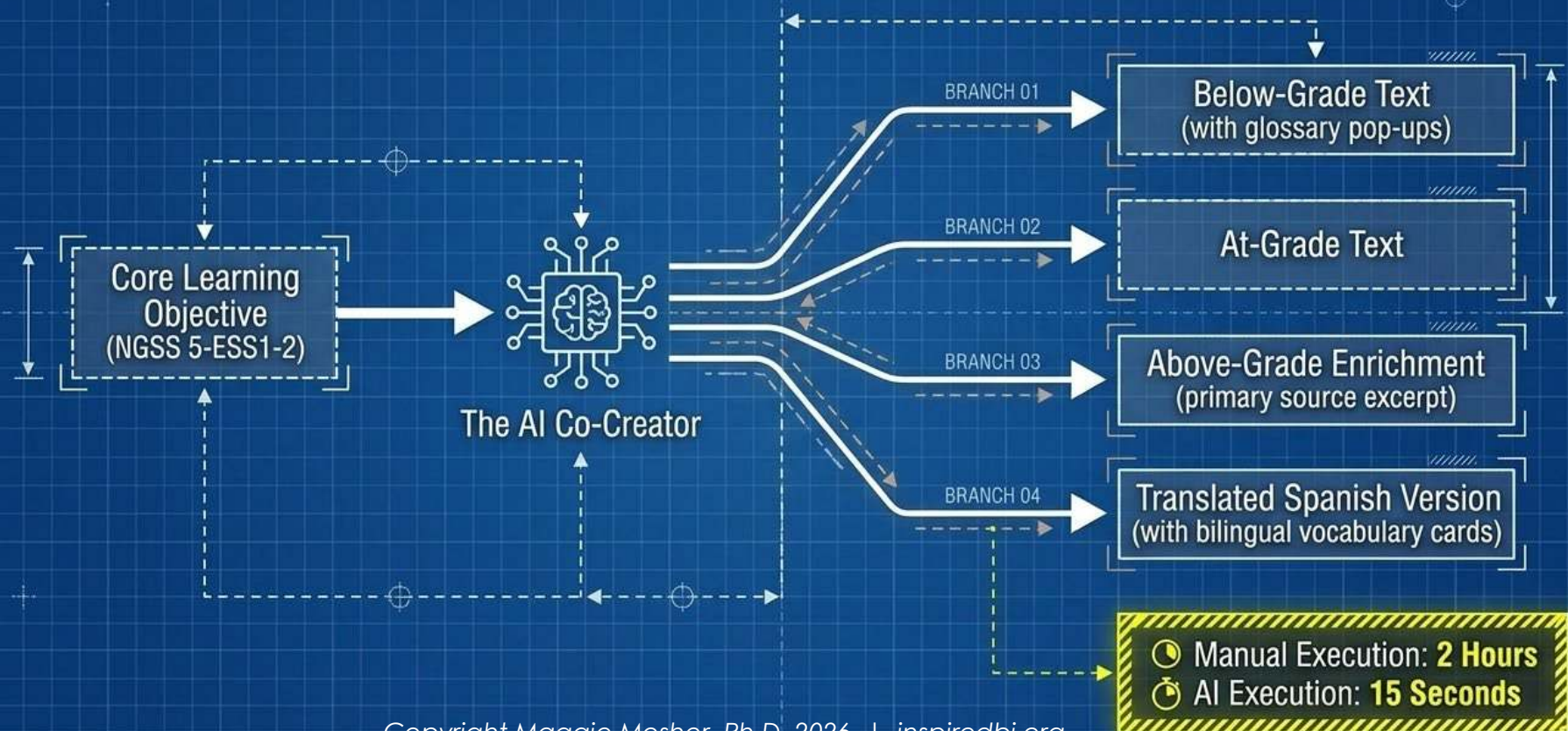


True 1:1 Personalization

Historically, true differentiated instruction was incredibly time-consuming. AI changes the equation.

Easily translate materials into multiple languages, adjust Lexile (reading) levels for the same text, or create alternative explanations for complex concepts.

Step 3: Mechanical Differentiation at Scale



10 AM TIME TO MOVE TO SAFETY

Why AI Integration Requires Immediate, Standardized Safety Protocols

Establish **clear, actionable safety protocols** to protect student privacy and academic integrity.

Key Focus Areas

- Safeguarding Student PII
- Preventing Algorithmic Bias
- Ensuring Regulatory Compliance



Federal Regulations: AI Data Governance

AI deployment in K-12 environments must strictly adhere to three primary federal regulations:



FERPA

Family Educational Rights and Privacy Act

Prohibits the unauthorized sharing of student educational records and PII with third-party AI vendors.



COPPA

Children's Online Privacy Protection Act

Requires verifiable parental consent for AI tools collecting personal data from children under 13.



CIPA

Children's Internet Protection Act

Mandates network filtering to block access to obscene or harmful AI-generated content on school networks.

Zero-Trust Data Privacy Protocols

Protect Student PII



PROACTIVE PROTECTION

Adopt Zero -Trust

Educators must adopt a zero-trust approach when interacting with external AI systems, assuming any unvetted platform could expose sensitive data.

Anonymize Data

Utilize anonymization techniques by replacing names with generic identifiers (e.g., '**Student A**') when analyzing classroom performance trends.



CRITICAL DIRECTIVE

Never input Student PII into public or unvetted AI generators.

Protected data includes names, ID numbers, behavioral records, IEP/504 details, and specific grades.

Multiple Interactive Modalities

- **Adaptive Learning:** Dynamically adjusts lesson difficulty based on real-time student performance.
- **Generative Chatbots:** Used for tutoring and drafting; pose risks of hallucination and data scraping.
- **Automated Assessment:** As s i s t s with grading and feedback, requiring oversight to ensure accuracy and fairness.



Algorithmic Bias Demands Continuous Educator Oversight

Audit for Algorithmic Bias

AI models inherit biases from their training data, which can disproportionately impact marginalized student groups (cultural, racial, gender).

Ensure Digital Equity

All students have equal access to premium, district-vetted AI tools.



Acceptable Use Policies (AUP) Must Explicitly Address Generative AI

1. Define Use

Define authorized student-AI interactions.

2. Set Boundaries

Differentiate AI assistance from academic dishonesty.

3. Prohibit Misuse

Ban malicious generation, filter bypasses, and impersonation.

4. Annual Sign -Off

Require annual review and signature from students and guardians.



ETHICS Framework

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E

Evaluate & Validate



Critically assess AI outputs, data sources, and the AI tools you choose to use.

T

Tool Application



When effectively implemented, ask: is it best for the job, or is human interaction better?

H

Human & Societal Impact



Consider the ethical and social effects of AI on people and communities.

I

Inner Workings & Fundamentals



Understand the foundational concepts of AI, such as how Large Language Models (LLMs) function.

C

Context & Governance



Maintain awareness of AI policy, accountability, and the context in which it is used.

S

Safety & Fairness

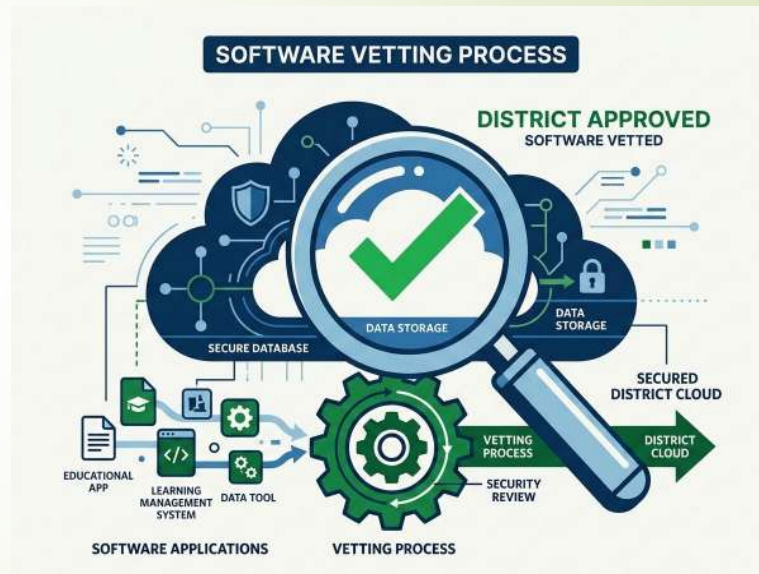


Ensure AI is responsible, respectful, and accessible for all.

Rigorous Procurement Procedures Prevent Unsafe Tool Adoption

Mandatory District Vetting Checklist

- Verify COPPA & FERPA compliance
- Review vendor data retention policies
- Confirm no student data is sold to third parties
- Ensure closed-loop training-free environment



All district -approved educational applications are funneled through a strict, multi -layered security and data privacy review.

AI Literacy Curricula

Develop Critical Student Competencies

Safety protocols are most effective when paired with comprehensive AI literacy education.

- **Understand Functions:** Teach students how AI models operate and process data.
- **Identify Biases:** Spot hallucinations and ethical implications of synthetic media.
- **Evaluate Outputs:** Treat AI generation as fallible drafts to foster digital citizenship.



Engage with AI 3: Evaluate whether AI outputs should be accepted, revised or rejected.

Knowledge: 3.1, 3.2 Skills: Critical Thinking Attitudes: Responsible

Basic

Learners understand why AI-generated outputs should be verified and recognise how they can verify the content themselves.

In the Classroom

Learners review an AI-powered tool's solution to a maths problem and accompanying explanation, then compare the AI-generated content to the process that their teacher has introduced to the class.



Intermediate

Learners verify AI-generated outputs for accuracy and relevance by consulting trusted sources and considering task-specific expectations.

In the Classroom

In small groups, learners plan an itinerary for a tourist visiting their hometown. When the teacher provides AI-generated recommendations for local restaurants, learners compare the recommendations using other sources to determine if the information is outdated, inaccurate or a "hallucination." From there, learners draw from their own knowledge and experience to decide whether the locations would be appealing to a visitor.

Advanced

Learners assess AI-generated outputs and justify decisions to accept, revise or reject them.

In the Classroom

When their teacher presents different AI-generated interpretations of a historical event, learners turn to trusted secondary or primary sources to verify the accuracy of the interpretations and argue which one they would include in a final report.

<https://ailiteracyframework.org/>

Manage AI 1: Decide whether to use AI systems based on the nature of the task.

Knowledge: 3.1, 4.3 Skills: Computational Thinking, Problem Solving Attitudes: Innovative, Responsible

Basic

Learners identify a variety of AI systems and the tasks they are designed to support.

In the Classroom

Learners follow guidance established by their teacher to match appropriate tools with academic tasks (e.g. AI for summarising, search engines for research).

Intermediate

Learners use what they know about AI to determine whether AI is an appropriate digital tool for a specific task.

In the Classroom

As the teacher calls out examples of everyday tasks for the whole class, learners move to corners of the room labelled "AI Only," "AI-Supported," "Human Only," or "Not Sure" based on how they think AI could be used. Learners then defend and explain their choices to peers who had chosen differently and switch corners if convinced.

Advanced

Learners determine whether AI is the right digital tool for a specific task by comparing the task's complexity and need for human judgement with the ethical implications of AI use.

In the Classroom

Learners consider the steps to write an essay (i.e. choosing a topic, researching evidence, organising arguments, drafting paragraphs, revising and proof-reading), then identify which steps AI could assist with and which steps require their own voice and reasoning. They consult their school's guidelines for AI use to tailor their approach.

Incident Response Plans

Address Policy Violations Swiftly

Clear incident response protocols must be activated in the event of AI misuse.

IT Response

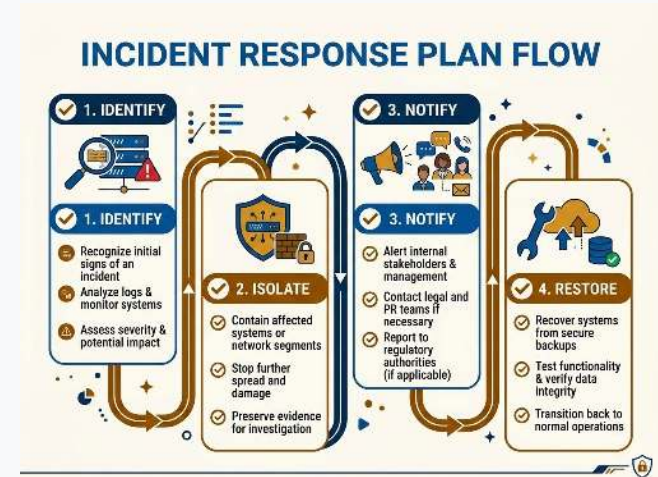
Data Breaches

If a data breach occurs via an unapproved AI tool, IT must immediately isolate the compromised account and notify affected families per legal requirements.

Student Guidance

Policy Violations

For student policy violations, such as generating malicious content or severe academic dishonesty, schools must follow a restorative justice framework combined with temporary revocation of AI access privileges.



Parent Collaboration Requires Transparency and Consent

DOCUMENTATION

Clear Disclosures

Schools must provide clear documentation outlining:

- Which AI tools are actively used
- The educational purpose of the technology
- Exactly what data is collected

CONSENT & SAFETY

Opt-Out & Trust

Schools must establish robust pathways for parent agency:

- Accessible opt-out procedures
- Verifiable COPPA-compliant consent
- Community sessions to address safety concerns



Professional Development: AI Integration

1. Safety Protocols

Mandatory professional development ensures all staff understand and can enforce AI safety protocols.

2. Core Training Modules

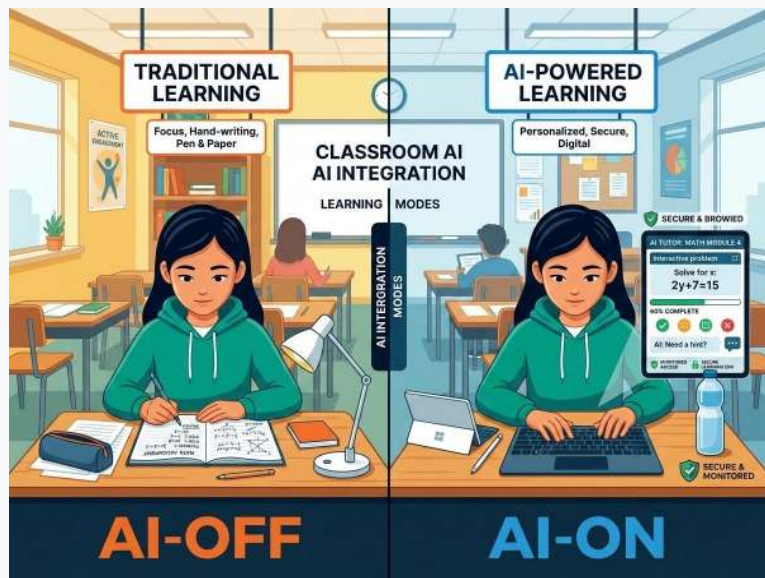
Training modules cover secure prompt engineering, AI detection, and integrating AI literacy into lessons.

3. Continuous Coaching

Continuous, job-embedded coaching keeps educators updated on rapidly evolving tools and district policies.



Classroom Operations Require Active Teacher Supervision



AI-On Mode

Dashboard Monitoring

Teachers utilize district-approved dashboard monitoring to ensure students remain on task and do not input inappropriate prompts.

Walled Gardens

Highly structured, walled-garden AI environments are used for elementary education, with complex tools restricted to older students.

AI-Off Mode

Foundational Skills

Instructional periods are established to balance technology-assisted learning with traditional, foundational skill development.

Age Gating

Strict operational procedures gate and restrict complex generative tools, safeguarding younger students' early development phases.

Academic Integrity Requires Transparent AI Citation Standards

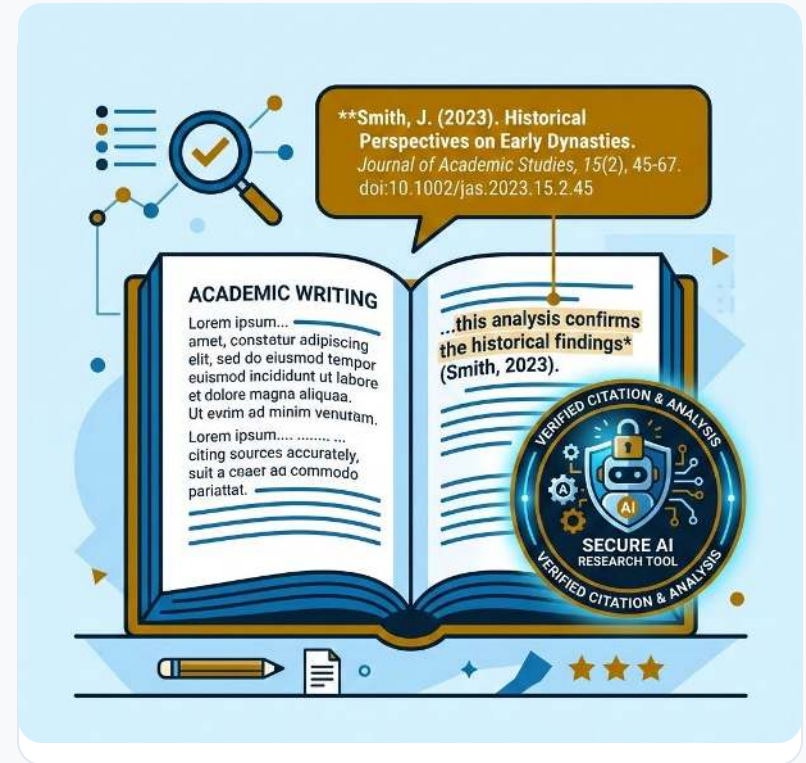
Process -Focused Evaluation

Students must transparently document their use of AI in academic submissions.

Educators must shift assessment strategies from product-focused grading to process-focused evaluations, utilizing oral defenses, in-class writing, and version history tracking to authenticate student understanding.

Academic Honesty Declaration Protocol

"Proper citation standards require students to declare which AI tool was used, the specific prompts entered, and how the generated output was modified in the final assignment."



Educator Safety Guidelines

Dictate Professional Boundaries

Approved Uses

- Generate rubrics
- Differentiate reading levels
- Brainstorm activities

Strict Limits

- **Never upload** student work into unapproved AI tools for automated grading (violates intellectual property and privacy standards)
- **Rigorously fact-check** all AI-generated lesson materials prior to instructional use



<https://ailiteracyframework.org/>



Threat Mitigation Defends Against Deepfakes & Phishing

Critical Protocol

Staff must be trained to verify urgent financial or data requests through secondary channels and report any suspected synthetic media to administration immediately to prevent social engineering attacks.

Network Security Enforces Safe AI Access Controls



Robust Content Filtering: Implement firewall rules to manage district-wide AI access and block high-risk domains.



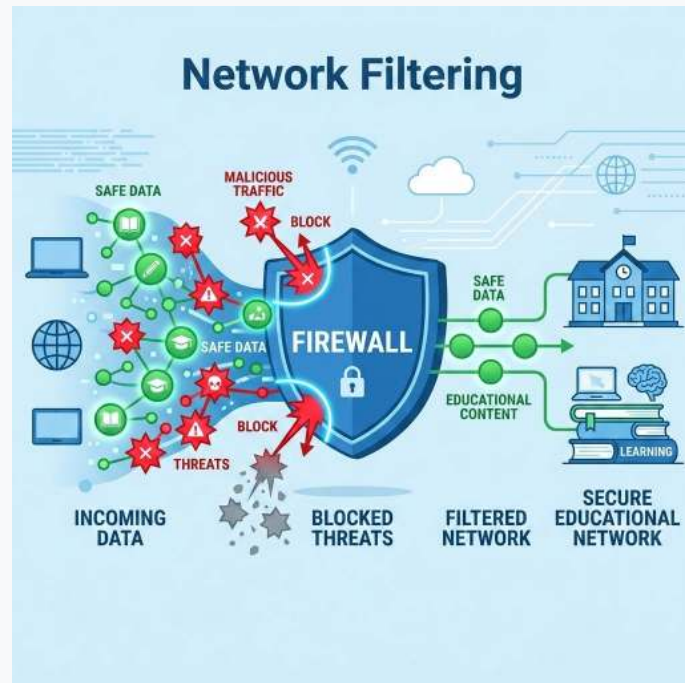
Role-Based Access Controls: Grant educators' broader administrative access while restricting student permissions.



SafeSearch Enforcement: Enforce strict search parameters on approved generative AI tools across the network.



Automatic Safety Filters: Restrict student interfaces to automatically block explicit, violent, or self-harm prompts.



Operationalizing Safety Protocols

- **Audit current practices:** No unvetted, personal AI accounts for school business.
- **Implement the AUP:** Ensure all students review and sign the updated AI Acceptable Use Policy.
- **Protect Data:** Adopt strict anonymization habits when engineering prompts.
- **Educate:** Integrate foundational AI literacy and critical evaluation skills into weekly plans.





Eduaide



SchoolAI



TeachAid



BriskAI



NapkinAI



SunoAI



Snorkel



Groovelit



Flint AI



AutoDraw



Ideogram



Padlet TA



Hailuo AI



Curipod



Question Well



Claude Code



Goblin Tools



FigJam Jambot



Formative AI



MirrorTalk



EduAide.ai



Hello History



EnlightenAI



Gamma



Antimatter Sorcerer



MetaGlasses

CHALKIE



Socrait



Grok

Wolfram Alpha

FlexClip



Microsoft Copilot

Reading Coach

Clipchamp

Humata

The Break-Down-Barriers Approach



Purposefully Selecting & Building AI Tools That Solve Real Classroom Problems

1

IDENTIFY

Name your #1 time-drain.

Be specific, not 'planning' but 'writing 30 differentiated exit tickets every Friday'.

2

EVALUATE

Is this repetitive or follow a pattern? Could another do it from my notes? If yes → AI candidate.

3

SELECT

Match the barrier to the right AI tool type: generative (creates content) vs agentic (takes action) and safety level. Don't over-engineer.

4

BUILD

Determine regulations and if using appropriately. If you are build it. Iterate weekly. Your AI gets smarter as you refine it together.

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Connect & Continue Learning

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Questions & Discussion — Let's Talk!



inspiredbi.org